

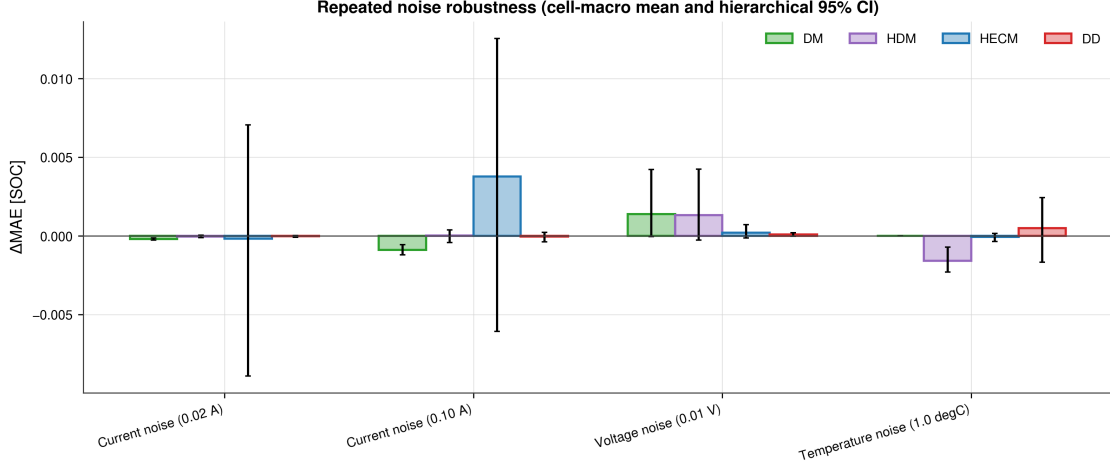


Figure 6.8: Repeated noise robustness across the holdout test set. Bars show cell-macro ΔMAE and error bars show hierarchical 95% confidence intervals with random seeds nested within cells. HECM is most affected by high current noise, while most other channel and model combinations remain close to baseline relative to the between-cell uncertainty.

Figures 6.7 and 6.8 together show that zero-mean sensor noise changes window-wide accuracy only slightly for most model and channel combinations. The local current-noise analysis nevertheless demonstrates that aggregate ΔMAE must be interpreted together with short-term output transmission, since rapid fluctuations can remain weakly represented by a window-wide metric.

6.4.5 Initialization error and recovery

The recovery experiment maps the same -0.10 SOC-equivalent mismatch, corresponding to ten percentage points, to the Coulomb-counting states in DM and HDM, the EKF SOC state in HECM, and a Q_c offset of $-0.10C_{\text{nom}}$ in DD. Figure 6.9 reports the paired trajectory differences from the common evaluation start, while recovery time remains referenced to the original intervention.

HECM provides the fastest persistent recovery with a cell-macro mean of 1.20 h, followed by DD at 2.60 h. DD enters the 2% recovery band temporarily in several runs but has a higher relapse fraction than HECM at 0.78 compared with 0.17. DM and HDM generally do not recover persistently within the observation horizon, and both have a persistent-censor fraction of 0.89. HECM also has the smallest excess-error area at 0.078 SOC h, compared with 0.116 SOC h for DD.

The observed recovery follows the available correction mechanisms. DM and